# Name the force and its source

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Identify forces on a chosen object and name each interaction partner.

**SUCCESS CRITERIA - I can...**

1. Name the chosen object and each force source.
2. Distinguish mass from weight with units.
3. Include only forces acting on the chosen object.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Velocity changes from -4 to -24 m/s in 4 s, then stays at -24 m/s for 8 s. Find acceleration and describe the speed trend in each interval.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $F_g = mg$.
Use g = 9.8 N/kg unless stated. Normal force is perpendicular to contact; tension follows the string. Name the source.

**Worked reasoning:** A book rests on a table. Earth pulls the book down; the table pushes the book up. Each force label names both the force and the object producing it.

**Your turn:** Draw the book's force diagram. Name the force on Earth paired with the book's weight.

**Marzano check - circle:** ① ② ③

## 03 Inventory the interactions

A 5 kg crate is pulled by a rope across a rough level floor. Identify every external force on the crate and the source of each. Do not assume force magnitudes without more information.



Add and label force arrows.

The dot represents the object. Chosen object: crate

**Marzano check - circle:** ① ② ③

## 04 Mass and weight in two places

A 3.0 kg object is on Earth (g=9.8 N/kg), then on the Moon (g=1.6 N/kg). Find each weight. What happens to mass?

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Normal is not always weight

A 5 kg box on a level floor has no vertical acceleration. A rope pulls with a 12 N upward component. Calculate the normal force and explain why it differs from mg.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A puck follows a circle at a steady 5 m/s. A student calls it equilibrium. Draw velocity at two points and correct the claim.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A 2.0 kg object hangs at rest from a vertical string. Name the two forces, their sources, magnitudes and directions. Is object-on-string a force on the object?

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 2 kg book rests on a table while a hand pushes down with 15 N. Use g=10 N/kg. Find the normal force and draw the third-law partner of each force on its separate object.

**Marzano check - circle:** 1 2 3